

Mathematics for Computer Science

COMP1021 - Durham University

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Part I

Module Content

11 Calculus

11.1 Introduction to Calculus

11.1.1 Sets

Definition 11.1.1.1: A set is an unordered collection of distinct elements denoted as $A = \{x | \text{condition}\}$.

Some common set operations include:

- **Definition 11.1.1.2:** The union of two sets, $A \cup B$, includes all the elements of both sets.
- **Definition 11.1.1.3:** The intersection of two sets, $A \cap B$, includes any elements which are present in both sets.
- **Definition 11.1.1.4:** If A and B are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (1)$$

- **Definition 11.1.1.5:** The set A is a subset of B , $A \subseteq B$, if every element of A is an element of B . A is a proper subset of B , $A \subset B$, if A is a subset of B and there is at least one element of B not in A .
- **Definition 11.1.1.6:** The powerset of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (2)$$

- **Definition 11.1.1.7:** The cartesian product of two sets, $A \times B$, is the set of all possible ordered pairs with the first element from A and the second from B .

Definition 11.1.1.8: For a finite set, the cardinality is the number of distinct elements that it contains.

11.1.2 Functions

For a function $f : A \rightarrow B$,

- **Definition 11.1.2.1:** A is known as the domain of f .
- **Definition 11.1.2.2:** B is known as the codomain of f .
- **Definition 11.1.2.3:** If $f(a) = b$, then b is the image of a .
- **Definition 11.1.2.4:** The pre-image of b is the set $\{a | f(a) = b\}$.
- **Definition 11.1.2.5:** f forms an injection if every element of A maps to a unique element in B , even if some elements of B don't have a mapping.
- **Definition 11.1.2.6:** f forms a surjection if every element of B maps to at least one element of A .
- **Definition 11.1.2.7:** f forms a bijection if it is both injective and surjective.

Definition 11.1.2.8: If for an unbounded set A , there is a bijective function $f : A \rightarrow \mathbb{N}$, then we say A is countably infinite, otherwise A is known as uncountably infinite.

11.1.3 Hyperbolic Functions

Definition 11.1.3.1: Given the unit hyperbola $x^2 - y^2 = 1$, the hyperbolic functions are defined by the coordinates of a point at which area $\frac{1}{2}\alpha$ has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

11.2 Multivariate Functions

11.2.1 Multivariate Derivatives

Definition 11.2.1.1: For a function $f(x, y)$, if we take $f = f_0$ to be fixed but unknown, then we can differentiate f as a function of x at $x = x_0$, giving us the partial derivative of f with respect to x which gives the gradient when moving in the x -direction from any point.

$$\frac{\delta f}{\delta x} [f(x, y)] = \lim_{x \rightarrow x_0} \left(\frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \right) = f_x \quad (3)$$

Definition 11.2.1.2: The vector of partial derivatives, denoted as ∇f , contains all the partial derivatives of a function. This forms a vector pointing in the direction of the greatest rate of increase of f , where its magnitude is the rate of increase in that direction. For example, for a function $f(x, y)$,

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (4)$$

Definition 11.2.1.3: The directional derivative of the function $f(x, y)$ gives the gradient at any point whilst looking in the direction given by the unit vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$.

$$\nabla_{\mathbf{v}} f = \lim_{h \rightarrow 0} \frac{f(x_0 + h\mathbf{v}) - f(x_0)}{h} \quad (5)$$

If the slope is approximately flat at a small scale, then the slope can be determined by

$$\nabla_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f \quad (6)$$

From this, we get that, for a function $f(\mathbf{x}_0)$ and a direction $\mathbf{v}$,

$$f(\mathbf{x}_0 + h\mathbf{v}) \approx f(\mathbf{x}_0) + h\mathbf{v} \cdot \nabla f(\mathbf{x}_0) \quad (7)$$

11.2.2 Multivariate Extrema

Definition 11.2.2.1: A function that is continuous is in class C^0 , and a function that is continuous and has a continuous derivative is in class C^1 . A function f such that f' , f'' , ..., $f^{(k)}$ exist and are continuous is in class C^k .

Definition 11.2.2.2: A function f such that derivatives of all orders exist and are continuous is in class C^∞ is also smooth.

Definition 11.2.2.3: All the second-order partial derivatives of a function $f(x, y)$ can be gathered into a Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (8)$$

This can be used to determine the form of bivariate extrema. Suppose $f(x, y) \in C^2$ and $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ then,

- (x_0, y_0) is a local maximum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} < 0$ at (x_0, y_0) .
- (x_0, y_0) is a local minimum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} > 0$ at (x_0, y_0) .
- (x_0, y_0) is a saddle point if $\det(\mathbf{H}_f) < 0$.
- If $\det(\mathbf{H}_f) = 0$, then the test is inconclusive.

At a stationary point x_0 , the quadratic approximation of a function $f(x)$ is

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} \mathbf{v}^T \mathbf{H}_f(x_0) \mathbf{v} \quad (9)$$

But as $\mathbf{H}_f(x_0)$ is real and symmetric, it has orthogonal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, so we can write

$$\mathbf{v} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + \dots + c_n \mathbf{e}_n \quad (10)$$

When substituted back into the original equation, this gives that

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} (\lambda_1(c_1)^2 + \lambda_2(c_2)^2 + \dots + \lambda_n(c_n)^2) \quad (11)$$

Given this, we can use the eigenvalues to determine the nature of the stationary point x_0 ,

- If all eigenvalues are positive, then the stationary point is a local minimum.
- If all eigenvalues are negative, then the stationary point is a local maximum.
- If there is a mixture of positive and negative eigenvalues, then the stationary point is a saddle point.

11.2.3 Vector-Valued Functions

Definition 11.2.3.1: A vector-valued function takes an n -dimensional vector as input, and outputs an m -dimensional vector, e.g., $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Definition 11.2.3.2: The Jacobian matrix of a vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be formed by setting $\mathbf{J}_{i,j} = \frac{\delta f_i}{\delta x_j}$ to form,

$$\mathbf{J}_f = \begin{pmatrix} \frac{\delta f_1}{\delta x_1} & \dots & \frac{\delta f_1}{\delta x_n} \\ \vdots & \ddots & \vdots \\ \frac{\delta f_m}{\delta x_1} & \dots & \frac{\delta f_m}{\delta x_n} \end{pmatrix} \quad (12)$$

For a multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix can be used to linearly approximate f at x ,

$$f(x + h\mathbf{v}) \approx f(x) + \mathbf{J}_f(x)(h\mathbf{v}) \quad (13)$$

11.3 Lagrange Multipliers

11.3.1 Constrained Optimisation

Definition 11.3.1.1: The goal of a constrained optimisation problem is to either minimise or maximise a multivariate function $f(\mathbf{x})$ where $x_1, x_2, \dots, x_n$ are not independent subject to some constraint $g(\mathbf{x})$.

Definition 11.3.1.2: The Lagrange multipliers method is used to find stationary points of an objective function $f(\mathbf{x}) \in C^1$ subject to the constraint $g(\mathbf{x}) = 0$.

1. Define a new objective function called the Lagrangian,

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x}) \quad (14)$$

2. Find the stationary points of $\mathcal{L}$ with respect to both $\mathbf{x}$ and λ , i.e., solve the system of equations $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$ and $g(\mathbf{x}) = 0$ for the unknowns $\mathbf{x}$ and λ .
3. Then, the constrained extrema are found among the solutions to these equations. Identify their nature.

11.4 Series

11.4.1 Introduction to Series

Definition 11.4.1.1: A sequence is an ordered, countably infinite, set of numbers.

Definition 11.4.1.2: A series is the sum of the members of a sequence.

Theorem 11.4.1.1: If the series $\sum_{n=0}^{\infty} a_n$ and the series $\sum_{n=0}^{\infty} b_n$ are convergent, and have respective sums α and β , then

$$\sum_{n=0}^{\infty} (a_n + b_n) = \alpha + \beta \quad (15)$$

and

$$\sum_{n=0}^{\infty} (a_n - b_n) = \alpha - \beta \quad (16)$$

Proof: $\sum_{n=0}^{\infty} a_n$ converges to α and $\sum_{n=0}^{\infty} b_n$ converges to β . Given an arbitrary $\epsilon > 0$, let $\epsilon_a = \frac{1}{2}\epsilon = \epsilon_b$. Since $\sum_{n=0}^{\infty} a_n$ converges, $\exists N_a$ such that,

$$\left| \sum_{r=0}^n a_r - \alpha \right| < \epsilon_a, \quad \forall n > N_a \quad (17)$$

Similarly, since $\sum_{n=0}^{\infty} b_n$ converges, $\exists N_b$ such that,

$$\left| \sum_{r=0}^n b_r - \beta \right| < \epsilon_b, \quad \forall n > N_b \quad (18)$$

Taking $N = \max\{N_a, N_b\}$, then $\forall n > N$,

$$\begin{aligned} \left| \sum_{r=0}^n (a_r + b_r) - (\alpha + \beta) \right| &= \left| \left(\sum_{r=0}^n a_r - \alpha \right) + \left(\sum_{r=0}^n b_r - \beta \right) \right| \\ &\leq \left| \sum_{r=0}^n a_r - \alpha \right| + \left| \sum_{r=0}^n b_r - \beta \right| \leq \epsilon_a + \epsilon_b = \epsilon \end{aligned} \quad (19)$$

i.e, $\sum_{n=0}^{\infty} (a_n + b_n)$ converges to $(\alpha + \beta)$ using triangle inequality $|a + b| \leq |a| + |b|$.

□

Theorem 11.4.1.2: If $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series $\sum_{n=0}^{\infty} a_n$ diverges.

Proof: Suppose $\sum_{n=0}^{\infty} a_n$ converges for some S , Then, $\lim_{n \rightarrow \infty} S_n = S$, which means that given arbitrary $\epsilon > 0$, there exists some N_ϵ such that, for all $n > N_\epsilon$,

$$|S_n - S| < \epsilon \quad |S_n - S| < 2\epsilon \quad (20)$$

Combining these,

$$2\epsilon > |S_n - S| + |S_{n+1} - S| = |S - S_n| + |S_{n+1} - S| \geq |S_{n+1} - S_n| = |a_{n+1}| \quad (21)$$

But ϵ was arbitrary, so we have shown that, for any $\epsilon' > 0$ there exists an N_ϵ such that $|a_n| < \epsilon'$ for all $n > N_\epsilon$.

□

As the number of elements in a series approaches infinity...

- **Definition 11.4.1.3:** If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.
- **Definition 11.4.1.4:** If $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.
- **Definition 11.4.1.5:** If $\lim_{n \rightarrow \infty} |a_n| = 0$, then the series absolutely converges.
- **Definition 11.4.1.6:** If $\lim_{n \rightarrow \infty} a_n = S$, then the series is convergent to S .

Theorem 11.4.1.3: The ratio test states that, if $\sum_{n=1}^{\infty} a_n$ is a series where $a_n \neq 0$ and $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ then

- a) If $L < 1$, the series absolutely converges.
- b) If $L > 1$, the series is divergent.
- c) If $L = 1$, then the test fails.

Proof: (a) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then $\exists r$ with $L < r < 1$ and N such that $\forall n > N$:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < r \quad \rightarrow \quad |a_{n+1}| < r|a_n| \quad (22)$$

Then $|a_{n+2}| < r|a_{n+1}| < r^2|a_n|$ and $|a_{n+3}| < r|a_{n+2}| < r^3|a_n|$ etc. So the remainder is

$$R_{n+m} = \lim_{m \rightarrow \infty} \left(|a_n| r^{m+1} \frac{1}{1-r} \right) = 0 \quad (23)$$

Hence the series converges. □

Definition 11.4.1.7: The n^{th} root test states that, for the series $\sum_{n=1}^{\infty} a_n$, $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$,

- If $L < 1$, then the series absolutely converges.
- If $L > 1$, then the series is divergent.
- If $L = 1$, then the test fails.

Definition 11.4.1.8: The alternating series test states that the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges if $a_n > 0$, $a_{n+1} \leq a_n \forall n$, and $\lim_{n \rightarrow \infty} a_n = 0$

Theorem 11.4.1.4: Every subsequence of a convergent sequence converges to the same limit as the original sequence.

Proof: Let $\{a_{n_k}\}$ be a subsequence of $\{a_n\}$. First note that $n_k \geq k$ and so $n_{k+1} \geq k+1$. Let $\lim_{n \rightarrow \infty} a_n = L$. Then, for any $\epsilon > 0$, there exists N_ϵ such that for all $n > N_\epsilon$ it holds that $|a_n - L| < \epsilon$. Let $k > N_\epsilon$, then $n_k > N_\epsilon$ and so it must be that $|a_{n_k} - L| < \epsilon$. □

11.4.2 Power Series

Definition 11.4.2.1: A power series is a series involving a variable x and a constant x_0 of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots \quad (24)$$

At any specific value of x , the power series becomes a normal series.

Definition 11.4.2.2: For an arbitrary power series $\sum_{n=0}^{\infty} a_n (x - x_0)^n$, we can use the ratio test (including the $(x - x_0)$ term). From this, we call $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r$ the radius of convergence. If $|x - x_0| < r$, we have absolute convergence and if $|x - x_0| > r$, we have divergence.

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $r > 0$. Then, within the interval $(-r, r)$:

- $f(x)$ is continuous.
- $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ (term-by-term differentiation).
- $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ (term-by-term integration).
- Both series $f'(x)$ and $\int_0^x f(t) dt$ have the same radius of convergence as $f(x)$.

Definition 11.4.2.3: Suppose we start with some function $f(x)$ and we wish to determine a power series $\sum_{n=0}^{\infty} a_n x^n$ such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Putting these values in, we see that if such a power series exists, then:

$$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + f^{(3)}(0)\frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} f^{(n)}(0)\frac{x^n}{n!} \quad (25)$$

This is called the Maclaurin series for f . To use the series, we must have that $f \in C^\infty$ and the power series converges.

Definition 11.4.2.4: We can recenter the power series on x_0 by defining a new function $g(x) = f(x + x_0)$ then $g(0) = f(x_0)$ and $g'(x) = f'(x + x_0)$. Repeatedly differentiating, $g^{(n)}(x) = f^{(n)}(x + x_0)$ so we build the Maclaurin series for g , which gives us

$$f(x + x_0) = g(x) = \sum_{n=0}^{\infty} g^{(n)}(0) \frac{x^n}{n!} = \sum_{n=0}^{\infty} f^{(n)}(x_0) \frac{x^n}{n!} \quad (26)$$

or,

$$f(x) = g(x - x_0) = f(x_0) + f'(x_0)(x - x_0) + f''(x_0) \frac{(x - x_0)^2}{2!} + \dots \quad (27)$$

This form is called the Taylor series expansion of f .

11.4.3 Taylor's Theorem

Lemma 11.4.3.1: If a real-valued function f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Proof: Not proved here.

□

Theorem 11.4.3.1: Suppose a function f is n times differentiable on $[a, x]$, then there exists some $\xi \in (a, x)$ such that

$$f(x) = f(a) + (x - a)f'(a) + \dots + \frac{(x - a)^{(n-1)}}{(n - 1)!} f^{(n-1)}(a) + R^n(x) \quad (28)$$

Where the remainder term is equal to $R_n(x) = \frac{(x-a)^n}{n!} f^{(n)}(\xi)$.

Proof: Suppose a function f is n times differentiable on $[a, x]$. Let k be a constant defined so that

$$f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!} f''(a) + \dots + \frac{(x - a)^{(n-1)}}{(n - 1)!} f^{(n-1)}(a) + \frac{(x - a)^n}{n!} k \quad (29)$$

We can then define the function

$$F(y) = f(x) - f(y) - (x - y)f'(y) - \dots - \frac{(x - y)^{(n-1)}}{(n - 1)!} f^{(n-1)}(y) - \frac{(x - y)^n}{n!} k \quad (30)$$

Then $F(x) = 0$ and also $F(a) = f(x) - f(a) = 0$ so we can apply Rolle's theorem to F on $[a, x]$, meaning there is some $\xi \in (a, x)$ such that $F'(\xi) = 0$.

□

11.4.4 Fourier Series

Definition 11.4.4.1: For a function $f(x)$, $-\pi < x < \pi$, the Fourier series is

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \quad (31)$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \quad (32)$$

12 Linear Algebra

12.1 Introduction to Linear Algebra

12.1.1 Mathematical Structures

Definition 12.1.1.1: Abstraction is the process of simplifying a complex problem via removing unnecessary details in order to apply more general problem solving techniques.

Definition 12.1.1.2: Groups consist of a set G and a binary operation $*$. They abide by the following axioms:

- **Definition 12.1.1.3:** Closure, when applying an operation between two elements of the group, the result will also be an element of the group.

$$\forall a, b \in G, a * b = c, c \in G \quad (33)$$

- **Definition 12.1.1.4:** Associativity, for any elements a, b, c in the group, the order in which the operation is performed does not matter, as long as the elements stay in their original order.

$$\forall a, b, c \in G, a * (b * c) = (a * b) * c \quad (34)$$

- **Definition 12.1.1.5:** Identity, the group contains an element such that when an operation is performed between an element in the group and the identity, the original element is returned.

$$\exists e \in G, \forall a \in G, a * e = e * a = a \quad (35)$$

- **Definition 12.1.1.6:** Inverse, for every element in the group, there exists an inverse element such that when an operation is performed between them, the identity is returned.

$$\forall a \in G, \exists b \in G, a * b = e \quad (36)$$

Definition 12.1.1.7: Fields contain a set F with two binary operations (addition and multiplication) and abide by the following axioms:

- Associativity (defined above).
- **Definition 12.1.1.8:** Commutativity - when applying an operation between two elements of the field, the order the elements appear in does not matter.

$$\forall a, b \in F, a + b = b + a \text{ and } a \times b = b \times a \quad (37)$$

- **Definition 12.1.1.9:** Identity, there exists two distinct elements, 0 and 1, in F such that,

$$\exists 0, 1 \in F, \forall a \in F, a + 0 = a \text{ and } a \times 1 = a \quad (38)$$

- **Definition 12.1.1.10:** Additive inverses, for every element in the field, there exists some other element, called the additive inverse, such that,

$$\forall a \in F, \exists b \in F, a + b = 0 \quad (39)$$

- **Definition 12.1.1.11:** Multiplicative inverses, for every element in the field, there exists some other element, called the multiplicative inverse, such that,

$$\forall a \in F, \exists b \in F, a \times b = 1 \quad (40)$$

- **Definition 12.1.1.12:** Distributivity of multiplication over addition. This axiom states that,

$$\forall a, b, c \in F, a \times (b + c) = (a \times b) + (a \times c) \quad (41)$$

Definition 12.1.1.13: A vector space is a mathematical structure with two sets, scalars F and vectors V , where V forms a commutative group under addition and F forms a field. They abide by the following axioms:

- Associativity (defined above).
- Identity (defined above).
- Distributivity of multiplication over addition (defined above).

12.1.2 Linear Combinations and Maps

Definition 12.1.2.1: Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ and scalars $a_1, \dots, a_n$, we can form a linear combination.

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \sum_{i=1}^n a_i\mathbf{v}_i \quad (42)$$

Definition 12.1.2.2: The set of all linear combinations of a set of vectors is called its span, denoted $\text{span}(V)$.

Definition 12.1.2.3: A set of vectors S is a spanning set for a vector space V if $\text{span}(S) = V$. We can equivalently say that S spans V .

Definition 12.1.2.4: A set of vectors S is linearly dependent if there exists a linear combination such that

$$\sum_{i=1}^n a_i\mathbf{s}_i = \mathbf{0} \text{ and } \exists i, a_i \neq 0 \quad (43)$$

Otherwise, the set is described as linearly independent.

Definition 12.1.2.5: A subset S of the vector space V is a basis for V if S is a spanning set and is linearly independent.

Theorem 12.1.2.1: For a vector space V , if S is a basis for V , then it must also be a minimal spanning set.

Proof: Let S be a basis of V . Assume that S is not a minimal spanning set, then there must be some element $s \in S$ such that $\text{span}(S \setminus \{s\}) = V$. Then, s can be written as a linear combination of elements from $S \setminus \{s\}$. That is, we have that

$$1 \cdot \mathbf{s} - (a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \dots + a_n\mathbf{s}_n) = \mathbf{0} \quad (44)$$

This means that S would be linearly dependent, contradicting the definition of S as a basis. Therefore, S must be a minimal spanning set.

□

Definition 12.1.2.6: The dimension of a vector space V is the cardinality of its basis S .

Definition 12.1.2.7: If we have a function $f : V \rightarrow W$ where V and W are vector spaces over a field F , then f is a linear map if,

$$\exists \mathbf{u}, \mathbf{v} \in V, \exists a \in F, f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \text{ and } f(a \cdot \mathbf{v}) = a \cdot f(\mathbf{v}) \quad (45)$$

These can also be called linear transformations or vector space homomorphisms.

Definition 12.1.2.8: If $V = W$ for a linear map f , then f is also a endomorphism.

12.2 Matrices

12.2.1 Matrix Operations

When the determinant of a matrix is zero, its columns are linearly dependent vectors. Therefore, if two columns are the same, then the determinant will be zero. Similarly, if a matrix A is identical to another matrix B , except for two columns being swapped, then $\det(A) = -\det(B)$.

Definition 12.2.1.1: To calculate a determinant, we use the Laplace expansion,

$$\det(C) = \sum_{j=1}^n (-1)^{i+j} c_{(i,j)} M_{(i,j)} \quad 0 < i \leq n, i \in \mathbb{N} \quad (46)$$

where $c_{(i,j)}$ is the coefficient of the term at (i,j) and $M_{(i,j)}$ is the minor at coordinates (i,j) .

Definition 12.2.1.2: Under a linear map $f : X \rightarrow Y$, $f(a\mathbf{x} + b\mathbf{y}) = af(\mathbf{x}) + bf(\mathbf{y})$, $f(X)$ is known as the column space and is the span of the columns. Its dimension is known as the column rank (this is also the number of linearly independent columns). Row space and rank can be defined similarly.

Definition 12.2.1.3: The rank of a matrix is its row/column rank after the matrix is put into RREF.

Definition 12.2.1.4: If $f : X \rightarrow Y$ is a linear map, then the kernel or null space of f is the set of vectors that map to zero. The dimension of the kernel is known as the nullity. The kernel forms a vector space where all linear combinations of kernel elements are in the kernel.

Definition 12.2.1.5: Given a linear map on vector spaces $T : V \rightarrow W$, if V has finite dimensions then $\text{Rank}(T) + \text{Nullity}(T) = \dim(V)$.

12.2.2 Systems of Equations and Gaussian Elimination

Definition 12.2.2.1: A finite set of linear equations is called a system of equations or a linear system.

- **Definition 12.2.2.2:** A solution is a sequence of numbers $s_1, s_2, \dots, s_n$ such that the assignment $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ satisfies every equation.
- **Definition 12.2.2.3:** A system is consistent if it has at least one solution.
- **Definition 12.2.2.4:** A system $A\mathbf{x} = \mathbf{b}$ is homogenous if $\mathbf{b}$ is a column of zeros. These systems have a trivial solution of $\forall x \in \mathbf{x}, x = 0$.

Definition 12.2.2.5: An augmented matrix is in row echelon form if the first non-zero number in each row is a one, any rows containing only zeros are grouped at the bottom, and if two successive rows are not all zeroes, then the leading one of the higher row is to the left of the leading one in the lower row.

Definition 12.2.2.6: An augmented matrix is in reduced row echelon form if it is in row echelon form and each column containing a leading one has zeroes everywhere else.

12.2.3 Geometry

Definition 12.2.3.1: Given a vector space X , a function $d : X \rightarrow \mathbb{R}$ is a norm if it satisfies:

- **Definition 12.2.3.2:** Triangular Inequality: $\forall x, y \in X, d(x + y) \leq d(x) + d(y)$.
- **Definition 12.2.3.3:** Absolute Homogeneity: $\forall x \in X, s \in \mathbb{R}, d(s \cdot x) = |s| \cdot d(x)$.
- **Definition 12.2.3.4:** Positive Definite: If $d(x) = 0$ then $x = 0$.

Definition 12.2.3.5: The Euclidean norm is the length of a vector and is given for $\mathbf{x} \in \mathbb{R}^n$ to be

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n (x_i)^2} \quad (47)$$

Any non-unit vector $\mathbf{x}$ can be used to define a unit vector $\hat{\mathbf{x}}$ in the direction of $\mathbf{x}$ by using

$$\hat{\mathbf{x}} = \frac{1}{\|\mathbf{x}\|} \mathbf{x} \quad (48)$$

Definition 12.2.3.6: The Manhattan distance is given by

$$d_M(\mathbf{x}) = \sum_{i=1}^n |x_i| \quad (49)$$

Definition 12.2.3.7: Given $p \in \mathbb{R}^+$, we can define the p-norm $\|\mathbf{x}\|_p$ on $\mathbb{R}^n$ as

$$\|\mathbf{x}\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad (50)$$

Using limits on the p -norm, we get

$$\lim_{p \rightarrow \infty} \|\mathbf{x}\|_p = \max(\mathbf{x}) \quad \lim_{p \rightarrow 0} \|\mathbf{x}\|_p = \infty \quad (51)$$

Definition 12.2.3.8: Two vectors are orthogonal if their dot product is zero.

Definition 12.2.3.9: Two vectors are normal if their Manhattan norm is one.

Definition 12.2.3.10: A set of vectors is orthonormal if they are all both orthogonal and normal.

12.3 Eigenvectors and Eigenvalues

12.3.1 Introduction to Eigenvectors and Eigenvalues

Definition 12.3.1.1: Let A be an $n \times n$ matrix. A non-zero vector $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$ is an eigenvector of A if, for some scalar λ ,

$$A\mathbf{x} = \lambda\mathbf{x} \quad (52)$$

Where λ is the eigenvalue of A corresponding to the eigenvector $\mathbf{x}$.

Definition 12.3.1.2: The equation $\det(\lambda I - A) = 0$ is called the characteristic equation of A . In general, this is a polynomial where n is in the order of A .

$$p(\lambda) = \lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \dots + c_n \quad (53)$$

Theorem 12.3.1.1: If A is an $n \times n$ matrix, then λ is an eigenvalue of A iff it satisfies $\det(\lambda I - A) = 0$.

Proof: By definition, λ is an eigenvalue of A iff $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$. We have

$$A\mathbf{x} = \lambda\mathbf{x} \iff A\mathbf{x} = \lambda I\mathbf{x} \iff (\lambda I - A)\mathbf{x} = \mathbf{0} \quad (54)$$

By theorem about invertible matrices, the last equation has solution $\mathbf{x} \neq \mathbf{0}$ iff $\det(\lambda I - A) = 0$.

□

Definition 12.3.1.3: For a matrix A with eigenvalue λ , the null space of $\lambda I - A$ is called its eigenspace where the non-zero vectors in this space are the eigenvectors corresponding to λ .

Let λ_0 be an eigenvalue of a matrix A :

- **Definition 12.3.1.4:** The algebraic multiplicity of λ_0 is the power k with which $(\lambda - \lambda_0)$ appears as a factor of $\det(\lambda I - A)$.
- **Definition 12.3.1.5:** The geometric multiplicity of λ_0 is the dimension of the eigenspace corresponding to λ_0 .

Definition 12.3.1.6: Principle component analysis is the process of identifying the principle components (most important eigenvectors) of a data set. It takes a data set in n dimensions and outputs an $n \times n$ matrix, demonstrating how each dimension varies with each other dimension (i.e., it captures the variance of one dimension of the data with another):

1. Pretend we have some covariance matrix $COV_{n \times n}$.
2. Find the n eigenvalues and corresponding eigenvectors of $COV_{n \times n}$.
3. Order the eigenvalues in decreasing order to get the importance of each.
4. Work out what percentage of the total value each eigenvalue contributes and take the top k values, contributing to 80% of the total.
5. Let $P_{n \times k}$ be a matrix with the k chosen eigenvectors as column vectors.

12.3.2 Complex Matrices

For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ and a scalar $k \in \mathbb{C}$, the following holds:

- $\overline{\overline{\mathbf{u}}} = \mathbf{u}$.
- $\overline{k\mathbf{u}} = \overline{k}\overline{\mathbf{u}}$.
- $\overline{\mathbf{u} + \mathbf{v}} = \overline{\mathbf{u}} + \overline{\mathbf{v}}$.
- $\overline{\mathbf{u} - \mathbf{v}} = \overline{\mathbf{u}} - \overline{\mathbf{v}}$.

If A is an $m \times k$ complex matrix and B is a $k \times n$ matrix, then:

- $\overline{\overline{A}} = A$.
- $\overline{A^T} = (\overline{A})^T$.
- $\overline{AB} = \overline{A}\overline{B}$.

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and a scalar $k \in \mathbb{C}$, the following holds:

- $\mathbf{u} \cdot \mathbf{v} = \overline{\mathbf{v}} \cdot \overline{\mathbf{u}}$.
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.
- $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$ and $\mathbf{u} \cdot (k\mathbf{v}) = \overline{k}(\mathbf{u} \cdot \mathbf{v})$.
- $\mathbf{v} \cdot \mathbf{v} \geq 0$ and $\mathbf{v} \cdot \mathbf{v} = 0$ iff $\mathbf{v} = \mathbf{0}$.

Theorem 12.3.2.1: If λ is an eigenvalue of a real $n \times n$ matrix A and $\mathbf{x}$ is a corresponding eigenvector, then $\overline{\lambda}$ is also an eigenvalue of A and $\overline{\mathbf{x}}$ is a corresponding eigenvector.

Proof: Since A is real, we have $A\overline{\mathbf{x}} = \overline{A\mathbf{x}} = \overline{\lambda\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}$. □

Theorem 12.3.2.2: If A is a real symmetric matrix, then all (complex) eigenvalues of A are real.

Proof: Let $\lambda \in \mathbb{C}$ be an eigenvalue of A and $\mathbf{x} \in \mathbb{C}^n$ a corresponding eigenvector. Take the complex conjugate of both sides of the equation $A\mathbf{x} = \lambda\mathbf{x}$. We get $\overline{A\mathbf{x}} = \overline{\lambda\mathbf{x}}$, and, since $A = \overline{A}$, it follows that $A\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}$. Then, using $A = A^T$, we compute the number $\overline{\mathbf{x}}^T A \mathbf{x}$ in two different ways:

$$\overline{\mathbf{x}}^T A \mathbf{x} = \overline{\mathbf{x}}^T (A\mathbf{x}) = \overline{\mathbf{x}}^T (\lambda\mathbf{x}) = \lambda(\overline{\mathbf{x}}^T \mathbf{x}) = \lambda(\mathbf{x} \cdot \mathbf{x}) = \lambda\|\mathbf{x}\|^2 \quad (55)$$

$$\overline{\mathbf{x}}^T A \mathbf{x} = (\overline{A\mathbf{x}})^T \mathbf{x} = (\overline{\lambda\mathbf{x}})^T \mathbf{x} = \overline{\lambda}(\overline{\mathbf{x}}^T \mathbf{x}) = \overline{\lambda}(\mathbf{x} \cdot \mathbf{x}) = \overline{\lambda}\|\mathbf{x}\|^2 \quad (56)$$

Since $\mathbf{x} \neq \mathbf{0}$, we have $\|\mathbf{x}\| \neq 0$, so $\lambda\|\mathbf{x}\|^2 = \overline{\lambda}\|\mathbf{x}\|^2$ implies $\lambda = \overline{\lambda}$, meaning $\lambda \in \mathbb{R}$. □

12.3.3 Inner Product Spaces

Definition 12.3.3.1: Let V be a real vector space. An inner product space on V is a function that associates to each pair $\mathbf{u}, \mathbf{v} \in V$ a real number $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$, satisfying the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $k \in \mathbb{R}$:

- **Definition 12.3.3.2:** The symmetry axiom: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
- **Definition 12.3.3.3:** The additivity axiom: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$.
- **Definition 12.3.3.4:** The homogeneity axiom: $\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$.
- **Definition 12.3.3.5:** The positivity axiom: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ iff $\mathbf{v} = \mathbf{0}$.

Definition 12.3.3.6: Vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V are orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Definition 12.3.3.7: Let W be a subspace in an inner product space V . Then the set

$$W^\perp = \{\mathbf{x} \in V \mid \langle \mathbf{x}, \mathbf{w} \rangle = 0 \forall \mathbf{w} \in W\} \quad (57)$$

is called the orthogonal complement of W .

Definition 12.3.3.8: Let $w_1, \dots, w_n \in \mathbb{R}^+$ be arbitrary numbers, which we'll call weights. The weighted Euclidean inner product on $\mathbb{R}^n$ is defined for vectors $\mathbf{u}, \mathbf{v}$ as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \dots + w_n u_n v_n \quad (58)$$

Definition 12.3.3.9: Let A be an invertible $n \times n$ matrix. Considering vectors in $\mathbb{R}^n$ as column vectors, define,

$$\langle \mathbf{u}, \mathbf{v} \rangle = A\mathbf{u} \cdot A\mathbf{v} \quad (59)$$

where the right-hand side uses the standard dot product in $\mathbb{R}^n$. This known as the inner product on $\mathbb{R}^n$ generated by A .

Definition 12.3.3.10: For vectors $\mathbf{p} = a_0 + a_1x + \dots + a_nx^n$ and $\mathbf{q} = b_0 + b_1x + \dots + b_nx^n$ in $\mathbb{P}_n$, define,

$$\langle \mathbf{p}, \mathbf{q} \rangle = a_0b_0 + a_1b_1 + \dots + a_nb_n \quad (60)$$

This is called the standard inner product on $\mathbb{P}_n$.

Definition 12.3.3.11: Fix distinct points $x_0, x_1, \dots, x_n \in \mathbb{R}$ (called sample points). For vectors $\mathbf{p} = p(x)$ and $\mathbf{q} = q(x)$ in $\mathbb{P}_n$, define

$$\langle \mathbf{p}, \mathbf{q} \rangle = p(x_0)q(x_0) + p(x_1)q(x_1) + \dots + p(x_n)q(x_n) \quad (61)$$

This is called the evaluation inner product at $x_0, x_1, \dots, x_n$.

Definition 12.3.3.12: Let $C[a, b]$ consist of all functions that are continuous on interval $[a, b]$. The operations in $C[a, b]$ are defined point-wise. For $\mathbf{f} = f(x)$ and $\mathbf{g} = g(x)$ in $C[a, b]$, define,

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_a^b f(x)g(x)dx \quad (62)$$

This is the inner product on the space $C[a, b]$.

Definition 12.3.3.13: For a complex vector $\mathbf{v}$, the Hermitian conjugation of $\mathbf{v}$, denoted $\mathbf{v}^\dagger$ is the conjugate transpose of the vector, i.e., $\mathbf{v}^\dagger = \overline{(\mathbf{v}^T)}$. A complex square matrix is a Hermitian matrix if it is equal to its own conjugate transpose.

Definition 12.3.3.14: Considering vectors in $\mathbb{C}^n$ as column vectors, the Hermitian inner product is defined as

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\dagger \mathbf{v} \quad (63)$$

Some standard (in)equalities for general inner products:

- **Definition 12.3.3.15:** Pythagoras' Theorem (inner-products), $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
- **Definition 12.3.3.16:** Cauchy-Schwarz Inequality (inner-products), $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$.
- **Definition 12.3.3.17:** Triangle Inequality (inner-products), $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

12.3.4 The Gram-Schmidt Process

Definition 12.3.4.1: A set of vectors in an inner product space is called orthogonal if all pairs of distinct vectors in the set are orthogonal. An orthogonal set consisting of unit vectors is called orthonormal.

Theorem 12.3.4.1: If $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthogonal basis in inner product space V , then for any $\mathbf{u} \in V$,

$$\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \dots + \frac{\langle \mathbf{u}, \mathbf{v}_n \rangle}{\|\mathbf{v}_n\|^2} \mathbf{v}_n \quad (64)$$

Moreover, if S is an orthonormal basis in V , then

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \dots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n \quad (65)$$

Proof: If $\mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$ then, since $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0$ for all $j \neq i$, we have that, for each i ,

$$\langle \mathbf{u}, \mathbf{v}_i \rangle = c_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle = c_i \|\mathbf{v}_i\|^2 \quad (66)$$

So $c_i = \frac{\langle \mathbf{u}, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2}$ as required. □

Definition 12.3.4.2: The Projection Theorem states that if W is a subspace in a finite-dimensional inner product space V , then every vector $\mathbf{u} \in V$ can be uniquely expressed as $\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$ where $\mathbf{w}_1 \in W$ and $\mathbf{w}_2 \in W^\perp$.

Definition 12.3.4.3: Every non-zero finite-dimensional inner product space V has an orthonormal basis. This can be found using the Gram-Schmidt process:

1. Let W be a subspace of V and let $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis of W .
2. Let $\mathbf{v}_1 = \mathbf{u}_1$. Clearly $\{\mathbf{v}_1\}$ is an orthogonal basis for $W_1 = \text{span}(\mathbf{u}_1)$.
3. If $\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}$ is an orthogonal basis for $W_{r-1} = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_N)$, find a vector $\mathbf{v}_r$ such that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is an orthogonal basis for W_r .

$$\mathbf{v}_r = \text{proj}_{W_{r-1}^\perp} \mathbf{u}_r = \mathbf{u}_r - \frac{\langle \mathbf{u}_r, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_r, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 - \dots - \frac{\langle \mathbf{u}_r, \mathbf{v}_{r-1} \rangle}{\|\mathbf{v}_{r-1}\|^2} \mathbf{v}_{r-1} \quad (67)$$

4. Since $\mathbf{v}_r \in W_{r-1}^\perp$, the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is orthogonal.
5. Once all vectors have been found, they can be normalised to form an orthonormal basis for V .

Theorem 12.3.4.2: If V is a finite-dimensional inner product space, then any orthogonal set of vectors in V can be extended into either an orthogonal or orthonormal basis.

Proof: Let $a = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthogonal set in V and $b = \{\mathbf{u}_{k+1}, \dots, \mathbf{u}_{k+n}\}$ some basis in V . Apply Gram-Schmidt to the b , and since a is an orthogonal set, we will have $\mathbf{v}_i = \mathbf{u}_i$ for $1 \leq i \leq k$. If $\mathbf{v}_r = \mathbf{0}$ at any step, do not add it to the output set. The final set will extend a , it will be orthogonal, and its span will be $\text{span}(a, b) = V$.

□

12.4 Decomposition

12.4.1 LU Decomposition

Definition 12.4.1.1: Matrix decomposition or factorisation is the process of representing a matrix as a product of two or more matrices with useful properties.

Definition 12.4.1.2: An elementary matrix is a matrix which differs from the identity matrix as a result of one elementary row operation (switching, addition, or multiplication).

Definition 12.4.1.3: LU decomposition is a method for solving linear systems by reducing the problem into two triangular systems. This method can be used on a square matrix A if U is its row echelon form, which was produced without any row exchanges being performed. The method is as follows:

1. Keep track of the row operations used to compute U by Gaussian elimination.
2. Let $E_1, \dots, E_k$ be the corresponding elementary matrices for each row operation.
3. Then the inverse matrices $(E_1)^{-1}, \dots, (E_k)^{-1}$ are easy to find.
4. Compute $L = (E_1)^{-1}(E_2)^{-1} \dots (E_k)^{-1}I$ by applying the corresponding row operations to the identity matrix.

Definition 12.4.1.4: A permutation matrix is a square matrix P obtained from I by permuting its rows.

Definition 12.4.1.5: A PLU decomposition of a square matrix A is a representation $A = PLU$ where P is a permutation matrix, L is lower triangular, and U is upper triangular.

12.4.2 Eigendecomposition

Definition 12.4.2.1: Eigendecomposition is the process of decomposing a matrix into PDP^{-1} form where P are eigenvectors and D is a diagonal matrix of eigenvalues.

Definition 12.4.2.2: Square matrices A and B are called similar if $A = P^{-1}BP$ for some invertible P . These matrices will have the same determinant, rank, trace, nullity, eigenvalues, etc.

Theorem 12.4.2.1: Matrices A and B are similar iff they represent the same linear operator $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, possibly in different bases.

Proof: Proof omitted.

□

Definition 12.4.2.3: A matrix A is called diagonalisable if it is similar to a diagonal matrix, i.e., there exists an invertible matrix P such that $P^{-1}AP$ is diagonal. Then P is said to diagonalise A .

Theorem 12.4.2.2: An $n \times n$ matrix is diagonalisable iff it has n linearly independent eigenvectors.

Proof: ($\implies$). Assume that there is an invertible matrix P and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that $D = P^{-1}AP$, or $AP = PD$. Denote the i column vectors of P as $\mathbf{p}_i$, then

$$AP = A[\mathbf{p}_1 | \dots | \mathbf{p}_n] = [A\mathbf{p}_1 | \dots | A\mathbf{p}_n] \quad (68)$$

On the other hand,

$$PD = [\lambda_1 \mathbf{p}_1 | \dots | \lambda_n \mathbf{p}_n] \quad (69)$$

Since $AP = PD$, we conclude that $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$ for all $1 \leq i \leq n$. Since P is invertible, its rank is n and so the vectors $\mathbf{p}_1, \dots, \mathbf{p}_n$ are linearly independent, making each of them an eigenvector. $\square$

($\impliedby$). Assume that A has n linearly independent eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_n$ and let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues. Define

$$P = [\mathbf{p}_1 | \dots | \mathbf{p}_n] \quad \text{and} \quad D = \text{diag}(\lambda_1, \dots, \lambda_n) \quad (70)$$

Then

$$AP = A[\mathbf{p}_1 | \dots | \mathbf{p}_n] = [A\mathbf{p}_1 | \dots | A\mathbf{p}_n] = [\lambda_1 \mathbf{p}_1 | \dots | \lambda_n \mathbf{p}_n] = PD \quad (71)$$

The columns of P are linearly independent, so its rank is n and it is invertible. Finally, $AP = PD$ is equivalent to $D = P^{-1}AP$. $\square$

To diagonalise a matrix A , we apply the following algorithm:

1. Find the eigenvalues of A .
2. Find a basis in each eigenspace of A and merge these bases into one set S .
3. If S has fewer than n vectors then A is not diagonalisable.
4. Else, form the matrix $P = [\mathbf{p}_1 | \dots | \mathbf{p}_n]$ where $S = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$. The set S contains n vectors and is linearly independent so is a basis for $\mathbb{R}^n$, hence P is invertible.
5. The matrix $D = P^{-1}AP$ is diagonal, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ where, for each i , λ_i is the eigenvalue corresponding to $\mathbf{p}_i$.

Definition 12.4.2.4: Let A be a diagonalisable matrix. We have $A = PDP^{-1}$. Since the factors in this decomposition are made of eigenvectors and eigenvalues, it is called an eigendecomposition of A .

Theorem 12.4.2.3: If vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are eigenvectors of a matrix A corresponding to pairwise distinct eigenvalues $\lambda_1, \dots, \lambda_k$, then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent.

Proof: Let $r \leq k$ be the largest number such that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent. Assume for contradiction that $r < k$, so $\{\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}\}$ is linearly dependent:

$$c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r + c_{r+1} \mathbf{v}_{r+1} = \mathbf{0} \quad (72)$$

where not all $c_1, \dots, c_r, c_{r+1}$ are 0. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent, we conclude that $c_{r+1} \neq 0$. Since $\mathbf{v}_{r+1}$ is an eigenvector, we conclude that $c_i \neq 0$ for some $i \leq r$. We assumed that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent, but $\{\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}\}$ is not. We derived that $c_{r+1} \neq 0$ and $c_i \neq 0$ for some $i \leq r$. Left multiply both sides of the linear combination by A and use $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$:

$$c_1 \lambda_1 \mathbf{v}_1 + \dots + c_r \lambda_r \mathbf{v}_r + c_{r+1} \lambda_{r+1} \mathbf{v}_{r+1} = \mathbf{0} \quad (73)$$

Instead multiplying the original equation by λ_{r+1} and subtracting the result, we get

$$c_1 (\lambda_1 - \lambda_{r+1}) \mathbf{v}_1 + \dots + c_r (\lambda_r - \lambda_{r+1}) \mathbf{v}_r = \mathbf{0} \quad (74)$$

So $c_i (\lambda_i - \lambda_{r+1}) = 0$, and hence $c_i = 0$, for all $i \leq r$, a contradiction. $\square$

12.4.3 QR Decomposition

Theorem 12.4.3.1: If A is an $m \times n$ matrix with linearly independent column vectors, then it can be factored as $A = QR$ where Q has orthonormal columns and R is an invertible, upper-triangular matrix.

Proof: Let A be an $m \times n$ matrix with linearly independent columns $\mathbf{u}_1, \dots, \mathbf{u}_n$. Let Q be the matrix formed from the column vectors given by the orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ obtained by applying Gram-Schmidt to $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. Or, in matrix form:

$$[\mathbf{u}_1 | \dots | \mathbf{u}_n] = [\mathbf{q}_1 | \dots | \mathbf{q}_n] \begin{pmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{q}_2 \rangle & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{q}_n \rangle & \langle \mathbf{u}_2, \mathbf{q}_n \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_n \rangle \end{pmatrix} = QR \quad (75)$$

From Gram-Schmidt, for each $j \geq 2$, $\mathbf{q}_j$ is orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$. Hence,

$$R = \begin{pmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \langle \mathbf{u}_n, \mathbf{q}_n \rangle \end{pmatrix} \quad (76)$$

□

Definition 12.4.3.1: The least squares problem states that, given a linear system $A\mathbf{x} = \mathbf{b}$ with m equations and n variables, find a vector $\mathbf{x}$ that minimises $\|\mathbf{b} - A\mathbf{x}\|$ (w.r.t. to the Euclidean inner product on $\mathbb{R}^n$). We call such a vector $\mathbf{x}$ a least squares solution to the system, the vector $\mathbf{b} - A\mathbf{x}$ is the least squares error vector, and the number $\|\mathbf{b} - A\mathbf{x}\|$ is the least squares error.

Theorem 12.4.3.2: If W is a finite-dimensional subspace in an inner product space V and $\mathbf{b} \in V$, then $\text{proj}_W \mathbf{b}$ is the best approximation to $\mathbf{b}$ from W in the sense that

$$\|\mathbf{b} - \text{proj}_W \mathbf{b}\| \leq \|\mathbf{b} - \mathbf{w}\| \quad (77)$$

For each vector $\mathbf{w} \in W$, and the inequality is strict for all $\mathbf{w} \neq \text{proj}_W \mathbf{b}$.

Proof: For any vector $\mathbf{w} \in W$, write

$$\mathbf{b} - \mathbf{w} = (\mathbf{b} - \text{proj}_W \mathbf{b}) + (\text{proj}_W \mathbf{b} - \mathbf{w}) \quad (78)$$

By Pythagoras' theorem, we have,

$$\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{b} - \text{proj}_W \mathbf{b}\|^2 + \|\text{proj}_W \mathbf{b} - \mathbf{w}\|^2 \geq \|\mathbf{b} - \text{proj}_W \mathbf{b}\|^2 \quad (79)$$

Moreover, the inequality is strict whenever $\mathbf{w} \neq \text{proj}_W \mathbf{b}$.

□

We can find a least squares solution by:

1. Let $W = \{A\mathbf{x} | \mathbf{x} \in \mathbb{R}^n\}$ be the column space for A .
2. Since $\text{proj}_W \mathbf{b}$ is the best approximation to $\mathbf{b}$ from W , the least squares solutions to $A\mathbf{x} = \mathbf{b}$ are exactly solutions to

$$A\mathbf{x} = \text{proj}_W \mathbf{b} \quad (80)$$

3. The representation $\mathbf{b} = \mathbf{w}_1 + \mathbf{w}_2$, $\mathbf{w}_1 \in W$, $\mathbf{w}_2 \in W^\perp$ is unique, so (80) is equivalent to the condition $\mathbf{b} - A\mathbf{x} \in W^\perp$.

4. The columns of A are the rows of A^T , so the condition can be rewritten as,

$$A^T A\mathbf{x} = A^T \mathbf{b} \quad (81)$$

Definition 12.4.3.2: The above equation is the normal equation associated with $A\mathbf{x} = \mathbf{b}$.

12.4.4 Linear Regression

Theorem 12.4.4.1: For any $m \times n$ matrix A , A has linearly independent columns iff $A^T A$ is invertible.

Proof: Both the columns of A being linearly independent and $A^T A$ being square are dependent on $A\mathbf{x} = \mathbf{0}$ only having one trivial solution. Each solution of $A\mathbf{x} = \mathbf{0}$ is a solution of $A^T A\mathbf{x} = \mathbf{0}$. Let $\mathbf{x}_0$ be a solution, then $A\mathbf{x}_0$ is both in the column space of A and in the null space of A^T , hence $A\mathbf{x}_0 = \mathbf{0}$. Thus $A\mathbf{x} = \mathbf{0}$ only has a trivial solution iff the same is true for $A^T A\mathbf{x} = \mathbf{0}$. □

To find a unique least squares solution, we let A be an $m \times n$ invertible matrix with linearly independent column vectors.

1. From the previous theorem, for every column $\mathbf{b} \in \mathbb{R}^m$, the least squares solution to $A\mathbf{x} = \mathbf{b}$ is unique:

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b} \quad (82)$$

2. For the last item, substitute $A = QR$ into the first equation above and simplify using $Q^T Q = I$ as Q has orthonormal columns.
3. If $A = QR$ is a QR-decomposition, then,

$$\mathbf{x} = R^{-1} Q^T \mathbf{b} \quad (83)$$

12.4.5 Spectral Decomposition

Definition 12.4.5.1: A square matrix Q is known as an orthogonal matrix if $Q^T = Q^{-1}$.

Theorem 12.4.5.1: For any $n \times n$ matrix Q , the following are equivalent:

1. Q is orthogonal.
2. The rows of Q form an orthogonal basis in $\mathbb{R}^n$.
3. The columns of Q form an orthogonal basis in $\mathbb{R}^n$.

Proof: For (1) $\iff$ (3): If $Q = [\mathbf{u}_1 | \dots | \mathbf{u}_n]$, then each entry (i, j) in the product $Q^T Q$ is $\langle \mathbf{u}_i, \mathbf{u}_j \rangle$. Hence, $Q^T Q = I$ iff we have $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $i \neq j$ and $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for all i , which holds iff the set $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is orthonormal. Any orthonormal set with n vectors in $\mathbb{R}^n$ is a basis. □

Properties of an orthogonal matrix:

- The transpose of an orthogonal matrix is also orthogonal.
- The inverse of an orthogonal matrix is also orthogonal.
- A product of orthogonal matrices is also orthogonal.
- If Q is orthogonal then $\det(Q) = \pm 1$.

Definition 12.4.5.2: Let A and B be two similar matrices such that $P^{-1}AP = B$ where B is also a diagonal matrix. If P can be chosen to be orthogonal, then A and B are orthogonally similar, and A is orthogonally diagonalisable.

Theorem 12.4.5.2: (The Spectral Theorem) For any $n \times n$ matrix A , the following are equivalent:

1. A is orthogonally diagonalisable.
2. A has an orthonormal set of n eigenvectors.
3. A is symmetric.

Proof: (1) $\iff$ (2) is the same as for general diagonalisation.

(1) $\implies$ (3), If $A = QDQ^T$, then $A^T = (QDQ^T)^T = (Q^T)^T D^T Q^T = QDQ^T = A$.

□

Theorem 12.4.5.3: If A is a symmetric matrix, then:

1. All eigenvalues of A are real.
2. Eigenvectors from different eigenspaces are orthogonal.

Proof: Proof for (1) is omitted.

(2) Assume that λ_1 and λ_2 are two different eigenvalues of A with corresponding eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$. We have:

$$\lambda_1 \langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle A\mathbf{u}_1, \mathbf{u}_2 \rangle \quad (84)$$

Expanding the dot product, we get that

$$(\mathbf{u}_1)^T A^T \mathbf{u}_2 = \langle \mathbf{u}_1, A\mathbf{u}_2 \rangle = \lambda_2 \langle \mathbf{u}_1, \mathbf{u}_2 \rangle \quad (85)$$

Since $(\lambda_1 - \lambda_2) \langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 0$ and $\lambda_1 \neq \lambda_2$, we get $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 0$ as required.

□

Orthogonally diagonalising a symmetric matrix:

1. Let A be a symmetric matrix.
2. Find the eigenvalues and a basis in each eigenspace of A .
3. Apply Gram-Schmidt to each basis to obtain an orthonormal basis for each eigenspace.
4. Form the matrix Q , whose columns are the vectors from Step 3. Q will orthogonally diagonalise A , and the diagonal $D = Q^T A Q$ contains the eigenvalues of A in the same order as the eigenvectors in Q .

Definition 12.4.5.3: Spectral decomposition: Let A be a symmetric matrix, $\mathbf{u}_1, \dots, \mathbf{u}_n$ be an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A , with $\lambda_1, \dots, \lambda_n$ the corresponding eigenvalues.

$$A = \lambda_1 \mathbf{u}_1 (\mathbf{u}_1)^T + \lambda_2 \mathbf{u}_2 (\mathbf{u}_2)^T + \dots + \lambda_n \mathbf{u}_n (\mathbf{u}_n)^T \quad (86)$$

Part II

References

13 Module References

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13.2 Cited Works

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